

## NumPy Practical

1. Create a 1D NumPy array of numbers from 1 to 20 and reshape it into a 4x5 matrix.
2. From a given array [10, 15, 20, 25, 30, 35], extract all values greater than 20.
3. Given two arrays:

```
a = np.array([1, 2, 3])
```

```
b = np.array([4, 5, 6])
```

Perform element-wise addition, subtraction, multiplication, and division.

4. Replace all odd numbers in the array below with -1
5. Reverse the following array using slicing:
6. Create a 4x4 matrix of numbers from 1 to 16. Extract the main diagonal elements.
7. Find the unique elements and their counts in the array:
8. Replace all values greater than 50 with 0 in this array
9. Find the index of the maximum value in the array
10. Given a 3x3 matrix, find the sum of each row and each column
11. Use NumPy to reshape the array below into a 2D array with 3 rows (use -1 for automatic column count)
12. Count how many values are divisible by 3 in this array
13. Sort each row of the following 2D array